

PSTAT 5LS Lab 4

Section 1

Announcements & Recap

Section 2

Learning Objectives

R Learning Objectives

- 1 Work with `pnorm()` and `qnorm()`
- 2 Introduce `prop_test()` to understand test statistics and confidence intervals

Statistical Learning Objectives

- 1 Understand how area under the normal curve relates to probability.
- 2 Understand how to move between probabilities and quantiles of the normal distribution.
- 3 Build intuition for the relationship between simulated p-values and p-values which arise from a normal approximation.
- 4 Understand the standard normal distribution and its corresponding z test statistic.
- 5 Understand how confidence intervals can provide an estimate for the true parameter.

Functions covered in this lab

- ① `plot_norm()`
- ② `pnorm()`
- ③ `qnorm()`
- ④ `prop_test()`

Section 3

Lab Tutorial

Pet Ownership Example

The American Pet Products Association found that a record high of 70% of all households in the United States had pets in 2020. A recent survey of 750 households found that 504 had at least one pet. Does the proportion of U.S. households with pets differ from 0.70?

Our hypotheses to test if there is a difference are

$$H_0 : p = 0.70 \text{ and } H_A : p \neq 0.70$$

where p represents the proportion of all households that have at least one pet.

The sample proportion is

$$\hat{p} = \frac{504}{750} = 0.672$$

Normal Theory Conditions

By the Central Limit Theorem (CLT), the sampling distribution of $\hat{p}$ will be approximately normal when we satisfy the following conditions:

- **Independence:** The observations are independent of one another.
- **Success-Failure:** We have at least 10 successes and 10 failures.
 - For hypothesis tests, the Success-Failure condition is met when $np_0 \geq 10$ and $n(1 - p_0) \geq 10$
 - For confidence intervals, the Success-Failure condition is met when $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$

Normal Theory Conditions

Let's check the conditions for this example:

- **Independence:** Since the 750 households are from a random sample, the observations will be independent of one another.
- **Success-Failure:** Since both $np_0 = 750(0.70) = 515$ and $np_0(1 - p_0) = 750(1 - 0.70) = 225$ are at least 10, the success-failure condition is satisfied.

Since both conditions are satisfied, the the sampling distribution of $\hat{p}$ will be approximately

$$N \left(\mu = 0.70, \sigma = \sqrt{\frac{0.70(1 - 0.70)}{750}} \right)$$

Visualizing the p-value

We can calculate the standard error (SE) and visualize where our observed $\hat{p} = 0.672$ falls on the normal curve.

Since we have a two-sided alternative hypothesis, we need to consider the other tail of the distribution as well.

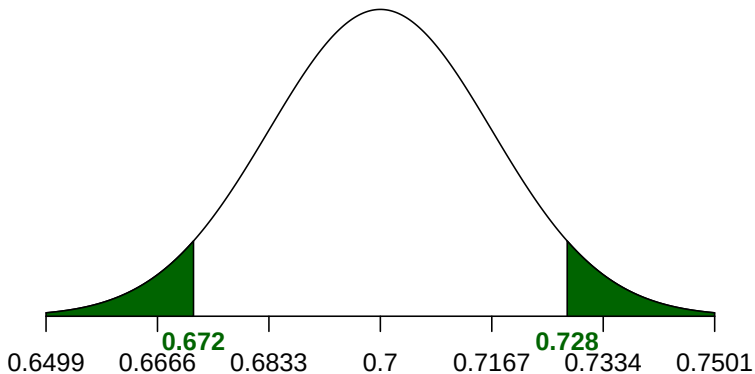
Run the `tryit1` code chunk in your notes to visualize the p-value.

```
SE <- round(sqrt(0.70 * (1 - 0.70) / 750),4)
```

```
plot_norm(mean = 0.70,  
          sd = SE,  
          shadeValues = c(0.672, 0.728),  
          direction = "beyond",  
          col.shade = "darkgreen")
```

Visualizing the p-value

N(0.7, 0.0167) Distribution



Approach 1: Using `pnorm()`

We can calculate the p-value (the area in the tails) by using `pnorm()`. Since we have a two-sided alternative, we find the area in one tail and then double it. Here, since $\hat{p}$ is less than p_0 , we find the area in the left tail and double it. Run the `tryit2` chunk in your notes to find the p-value.

```
2 * pnorm(q = 0.672, mean = 0.70, sd = 0.0167, lower.tail = TRUE)
```

Approach 1: Using `pnorm()`

The p-value is estimated to be

```
2 * pnorm(q = 0.672, mean = 0.70, sd = 0.0167, lower.tail = TRUE)
```

```
## [1] 0.09361159
```

Approach 2: Using `prop_test()`

The `pstat5lsSBI` package we have been using also contains a function called `prop_test()` that will calculate the value of the test statistic and the p-value for us. Professor Miller showed you this function briefly in lecture.

You will need to send the function the following arguments:

- `x`: the number of “successes” in the sample
- `n`: the sample size
- `p`: the hypothesized population proportion
- `alternative`: where to shade (`"two.sided"`, `"less"`, `"greater"`)
- `conf.level` (optional): used to get a confidence interval (we will use this later)

Using `prop_test` for Our Pet Example

We are testing $H_0 : p = 0.70$ and $H_A : p \neq 0.70$. You will need to enter values for

- `x` the number of successes (here the number of households with a pet)
- `n` the sample size
- `p` the value we assume p to be when H_0 is true
- `alternative` the direction of H_A ("two.sided", "less", "greater" – make sure the direction is in quotes)

in the `tryIt3` chunk in your notes before you run it.

```
prop_test(x = 504,  
          n = 750,  
          p = 0.70,  
          alternative = "two.sided")
```


Using prop_test for Our Pet Example

```
##  
## 1-sample proportions test without continuity correction  
##  
## data:  x out of n, null probability p  
## Z = -1.6733, p-value = 0.09426  
## alternative hypothesis: true p is not equal to 0.7  
## 95 percent confidence interval:  
##  0.6384 0.7056  
## sample estimates:  
##      p  
## 0.672
```

Comparing the p-values

To recap, we have *three* ways to compute the p-value for a one proportion hypothesis test:

- ❶ (Did not do this today) Create a set of simulated proportions using `simulate_chance_model()`, then using the histogram of these simulated proportions to find the p-value
- ❷ Compute the μ and σ for the approximate normal distribution and then use the `pnorm()` function.
- ❸ Use the `prop_test()` function (which also uses normal theory)

Each of these will produce a slightly different result. **No need to worry about how close the values should be, or which value is “best”.**

Conclusion

Since the p-value of 0.094 is greater than $\alpha = 0.05$, we fail to reject the null hypothesis. We don't have enough evidence to suggest that the proportion of households with pets has changed from the 70% during the pandemic.

From Testing to Estimation

We use confidence intervals when we

- are interested in estimating the population proportion (and not in testing a specific value)
- we have rejected H_0 in a hypothesis test and would like to find reasonable values for the population proportion.

Conditions for a Confidence Interval

As with hypothesis tests, we need to check both the **Independence** and **Success-Failure** conditions to determine if we can use the normal distribution to model the sampling distribution of $\hat{p}$.

The change for confidence intervals is that we check the Success-Failure condition using $\hat{p}$:

Success-Failure: Both $n\hat{p}$ and $n(1 - \hat{p})$ must be at least 10.

Approach 1: Calculating the CI by Hand

A confidence interval for p is

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where z^* is the **critical value**. For CIs for proportions, we have a handy table of common critical values

Confidence Level	z^*
90%	1.645
95%	1.960
98%	2.326
99%	2.576

Example: The “Invisible Gorilla” Study

In a famous study, participants watch a video of people passing a basketball and are told to count the number of passes. During the video, a person in a gorilla suit walks across the court, stops to beat their chest, and walks off.

In a recent replication of this experiment with a group of 150 college students, only 66 students noticed the gorilla. Use these results to calculate a 98% confidence interval for the population proportion of college students who would notice the gorilla.

Example: The “Invisible Gorilla” Study

First, we check the conditions. Think about the conditions and talk about them with a neighbor. Do you think the conditions are met?

Example: The “Invisible Gorilla” Study

Checking the conditions:

- **Independence:** We were not told that this is a random sample of students. However, the students did not know the purpose of the study, so it's reasonable to think that whether or not a student noticed the gorilla was independent of whether or not any other student noticed the gorilla. (Note that you might have a different opinion! That's okay, just explain your reasoning.)
- **Success-Failure:** Since $n\hat{p} = 66$ and $n(1 - \hat{p}) = 150 - 66 = 84$ are both greater than 10, this condition is satisfied.

Example: The “Invisible Gorilla” Study

To calculate the 98% CI by hand, we need to

- Calculate $\hat{p}$
- Identify z^*
- Calculate the SE
- Calculate the interval

Try this on your own, and we will come back together to compare answers.

Approach 2: Using `prop_test()`

The output from `prop_test()` also provides a confidence interval for the population proportion. By default, R gives us a 95% confidence interval. We need to specify `x` (number of successes), `n` (sample size), and the confidence level if we want something other than a 95% CI.

In the `tryIt4` chunk, enter `x`, `n`, and set `conf.level` to 0.98 to get a 98% confidence interval.

```
prop_test(x = 66,  
          n = 150,  
          conf.level = 0.98)
```

Using `prop_test()` to Find Confidence Intervals

```
##  
## 1-sample proportions test without continuity correction  
##  
## data:  x out of n, null probability 0.5  
## Z = -1.4697, p-value = 0.1416  
## alternative hypothesis: true p is not equal to 0.5  
## 98 percent confidence interval:  
##  0.3457135 0.5342865  
## sample estimates:  
##      p  
## 0.44
```

We estimate (at the 98% confidence level) that the population proportion of college students who would notice the gorilla is between 0.346 and 0.534.

Using `prop_test()` to Find Confidence Intervals

Caution: To get a two-sided confidence interval, the `alternative` argument *must* be set to `two.sided`. If it isn't, you will get a *confidence bound*.

Note that `two.sided` is the default for `prop_test()`, so if you just want a confidence interval you can leave the `alternative` argument off.

Confidence bounds can be useful when we have one-sided hypothesis tests, but we will leave them to your later statistics courses.

One More Example

X, the social media platform formerly known as Twitter, saw a name change and policy changes once it was under new ownership. Shortly after the name change, seventy-five percent (75%) of all X users said that they planned to keep using X in 2024. Researchers at X would like to know if this rate differs for Millennials, those born between 1981 and 1996. A random sample of 100 Millennial users of X revealed that 65% planned to keep using X in 2024.

What Would We Expect to See in the Sample?

If the rate of current Millennial users of X is the same as the rate for all X users, how many Millennial users of X from the sample of 100 would we expect to say that they planned to keep using X in 2024?

Incorrect Alternative

The researchers at X wanted to know if the rate of Millennials who planned to continue using X in 2024 differs from the rate of all users of X. The researcher in charge accidentally ran a one-sided hypothesis test and got the following output

```
prop_test(x = 65, n = 100, p = 0.75, alternative = "less")
```

```
##  
## 1-sample proportions test without continuity correction  
##  
## data:  x out of n, null probability p  
## Z = -2.3094, p-value = 0.01046  
## alternative hypothesis: true p is less than 0.75  
## 95 percent confidence interval:  
##  0.0000000 0.7284545  
## sample estimates:  
##      p  
## 0.65
```


Correcting the Mistake

The value of the test statistic for the incorrect alternative was $z = -2.309$. What should the test statistic be for the correct two-sided test?

- It should be half as much; that is, $z = -1.155$.
- It should be positive instead of negative; that is, $z = 2.309$.
- It should be the same; that is, $z = -2.309$.
- It should be twice as much; that is, $z = -4.618$.

Correcting the Mistake

The p-value for the incorrect alternative was 0.01046. What should the p-value be for the correct two-sided test?

- It should be half as much; that is, $p\text{-value} = 0.00523$.
- It should be negative instead of positive; that is, $p\text{-value} = -0.01046$.
- It should be the same; that is, $p\text{-value} = 0.01046$.
- It should be twice as much; that is, $p\text{-value} = 0.02092$.